

Nathanael Chwojko-Srawley

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EDUCATION

PhD at the University of Toronto <i>Doctorate in Mathematics</i>	SEPT 2025 - AUG 2029
Master Degree at the University of Toronto <i>M.Sc. in Mathematics. Thesis: the Hirzebruch–Riemann–Roch theorem</i>	SEPT 2024 - OCT 2025
Undergraduate Degree at the University of Toronto <i>Specialist in Mathematics, Minor in Computer Science. Conferred with High Distinction</i> Cumulative GPA 3.75/4.00. Annual GPA 4.00 in each of the final two years (2021 Summer through 2023 Winter). Dean’s List Scholar: 2019, 2021, 2022.	SEPT 2017 - JUNE 2023

RESEARCH INTERESTS

Algebraic geometry and number theory, and the topological invariants that detect algebraicity: Hodge theory and the integral Hodge conjecture, generalized cohomology (K-theory, twisted K-theory, algebraic cobordism), and Riemann–Roch type theorems. Secondary interests in probability and its applications, particularly measure-theoretic probability, stochastic processes, and statistical methodology for dependent data.

RESEARCH EXPERIENCE

Twisted K-theory and the Integral Hodge Conjecture  <i>Supervisor: Daniel Litt</i> <i>First-year Ph.D. research project</i> Asked whether twisted K-theory strengthens the topological obstruction to the integral Hodge conjecture. I axiomatized the possible factorizations of the cycle class map and proved that the twisted Atiyah–Segal d_3 differential agrees with the untwisted one, so the twisted refinement yields no new obstruction. Supporting work covered complex orientations via Panin–Smirnov, computations against Tripathy’s examples, and Levine–Morel algebraic cobordism.	SEPT 2025 - AUG 2026
Hirzebruch Riemann Roch Theorem  <i>Supervisor: Jacob Tsimerman</i> <i>Masters Thesis</i> Building off the classical results generalizing the Riemann Roch Theorem throughout the 20th century, this paper is an exposition on Hirzebruch’s methods translated into the 21st century. It includes a proof of Hirzebruch Riemann Roch and a synopsis of its algebraic generalization. The most recent developments of the generalization of the theorem are presented in the paper as well.	SEPT 2024 - AUG 2025
Parametric Dehn’s Lemma <i>Supervisor: Yevgeniy Liokumovich</i> <i>Received Grant of \$7500 From the MCSRA</i>	MAY 2023 - AUG 2023

Michael Freedman observed that the Parametric version of Dehn's Lemma is equivalent to the Poincare Conjecture. Using Classical techniques from Differential Geometry, I pushed further on finding isometries in the special case where there are generically at most 2 intersection points

Almost Global Existence for Nonlinear Wave Equations

Supervisor: Yakov Shlapentokh-Rothman

NSERC Undergraduate Student Research Award (USRA)

Building off of the work of Sergiu Klainerman, I worked on finding almost global existence for nonlinear wave equations in the setting of time dependent inhomogeneous media and Schwarzschild black hole exterior. Since developed into *Nonlinear Wave and Dispersive PDE*, an 87-page volume of the EYNTKA series (linked above).

MAY 2022
- AUG 2022

Classifying Automorphism-Representable Groups

Supervisor: Ehsaan Hossain

Research Opportunity Program (ROP)

Generalizing the classical result from group theory that there does not exist a group G such that $\text{Aut}(G) = \mathbb{Q}$, I worked on finding conditions on a group G for which $\text{Aut}(G) = H$ exists for various of groups H

MAY 2021
- AUG 2021

EXPOSITORY WRITING

Everything You Need To Know About (EYNTKA) series

Self-authored expository texts in pure and applied mathematics, each taking a reader from foundations to research-adjacent material. **44 volumes are published at <https://nathanaelsrawley.com/books>, totalling 15,898 pages**, 41 of them over 100 pages. The longest:

AUG 2019 -

- [Algebra](#) (2,053 pp.)
- [Modern Algebraic Geometry](#) (1,489 pp.)
- [Linear Algebra](#) (692 pp.)
- [Lie Groups](#) (590 pp.)
- [Elliptic Curves](#) (561 pp.)
- [Algebraic Topology](#) (504 pp.)
- [Riemannian Geometry](#) (492 pp.)
- [The Langlands Program for \$GL\(2\)\$](#) (450 pp.)
- [Abelian Varieties](#) (431 pp.)
- [Moduli and Stacks](#) (429 pp.)
- [Analytic Number Theory](#) (414 pp.)
- [Hodge Theory](#) (413 pp.)
- [Probability](#) (339 pp.)
- [Intersection Theory](#) (334 pp.)
- [Functional Analysis](#) (310 pp.)
- [Galois Cohomology](#) (246 pp.)

TALKS I HAVE GIVEN

Introduction to Modern Algebraic Geometry

Mathematics Seminar

I presented on the foundations of modern Algebraic Geometry, its connection to Complex Geometry and GAGA-like connections, as well as preliminary ideas on the classification of algebraic surfaces.

OCT 2022

TMWYF Insiders Perspective on Undergraduate Research

Speaker

I shared my experience doing math research, link can be found here:
[TMWYF Talk Link](#)

OCT 2022

Functional Analysis and Sobelev Spaces <i>Canadian Undergraduate Mathematics Conference (CUMC)</i> I presented on topics within Functional Analysis concerning Sobelev spaces, as well as their application to my research in PDEs	JULY 2022
Consequences of Nakayama's Lemma <i>Undergraduate Seminar</i> I presented Nakayama's lemma as well as some consequences (ex. a finitely generated module M over a commutative ring A implies that surjective endomorphisms are injective, and some local-to-global results)	Nov 2021
Applications of Category Theory <i>Undergraduate Seminar</i> I presented on the intuitions behind Category theory and 'Categorical thinking', common examples of categorical constructions, and the applications of Category theory to the fields of Algebra, Analysis, Physics, and Computer Science	Nov 2021
Automorphism-Representable Groups <i>Smarti Gras Convention</i> I presented my ROP research results, from elementary results using elementary group theory to some more recent developments involving methods using spectral sequences.	AUG 2021
Classification of Divisible Groups <i>Undergraduate Seminar</i> I presented on the classification of divisible groups, and elementary theory of injective modules.	JUL 2021

AWARDS AND GRANTS

NSERC Undergraduate Student Research Award (USRA) Awarded for <i>Almost Global Existence for Geometric Nonlinear Wave Equations</i> , supervised by Yakov Shlapentokh-Rothman.	2022
MCS Research Award (MCSRA) \$7,500 for summer research on parametric Dehn's lemma, supervised by Yevgeniy Liokumovich.	2023
Dean's List Scholar, University of Toronto	2019, 2021, 2022
Research Opportunity Program (ROP) Competitive undergraduate research placement, MAT499Y5, graded A+ (100).	2021

TEACHING EXPERIENCE

Head Teaching Assistant Coordinated the teaching-assistant team for first-year calculus cohorts of roughly 1250 students. Ran TA training, wrote and vetted tutorial worksheets and assessment questions, set grading standards and turnaround, fielded TA and student escalations, and acted as the liaison between the TA team and the course coordinators. Courses I was Head TA for are: - Calculus I for Life Sciences (MAT130) Sept 2026- - Calculus I (MAT135) Sept 2025-Dec 2025	SEPT 2025 - PRESENT
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Teaching Assistant (TA)

Prepared and ran tutorials, graded assignments, held office hours, and invigilated tests. Courses I was a TA for are:

- Linear Algebra II (MAT224)
Jan 2026-Apr 2026
- Calculus II (MAT136)
Jan 2026-Apr 2026
- Abstract Mathematics (MAT246)
Jan 2025-Apr 2025
- Linear Algebra I (MAT223)
Sept 2024-Dec 2024
- Calculus I (MAT135)
Sept 2024-Dec 2024
- Abstract Algebra (MAT301)
Jan 2023-Apr 2023
- Abstract Algebra (MAT301)
Sept 2021-Dec 2021
- Linear Algebra II (MAT224)
July 2021-Aug 2021
- Linear Algebra II (MAT224)
May 2021-June 2021
- Combinatorics (MAT202)
Jan 2021-Apr 2021
- Calculus II (MAT136)
Sept 2020-Dec 2020

SEPT 2021

- PRESENT

Math Learning Centre

Drop-in tutoring for first- and second-year mathematics courses, working with students one-on-one across the full range of MAT1xx/MAT2xx material.

SEPT 2026 -

Guest Lecturer, MAT135

Asked the course instructor for lecture experience, then prepared the material, organised the lecture-TA duties, and taught the class. Surveyed students afterwards on pacing and clarity and received strongly positive feedback.

2024

SOFTWARE

Open-source Neovim ecosystem

Author and maintainer of 11 published plugins (33 GitHub stars), mostly LaTeX and reference-management tooling for academic writing: citation and label pickers, a shared navigation backend, and context-dispatched editing. Also the NoetherVim distribution, and five fixes merged upstream in vimtex, vim-matchup, telescope-zotero.nvim and workspace-diagnostics.nvim. <https://github.com/Chiarandini>

2023 -

Research software

Market microstructure engine (Rust, Python): limit order book reconstruction from raw exchange feeds, 282M messages/session, with falsifiable consistency checkers and a multiple-testing-corrected study of order-flow imbalance. *Path signatures* (Rust): the signature transform implemented from the tensor algebra up, with Chen's identity and a Lyndon-word basis of the free Lie algebra. *collatz-chains* (Rust): symbolic Monte Carlo over the accelerated Collatz map.

2025 -

Trilingual dictionary

English/Japanese/Chinese dictionary using triangular translation through an English pivot. Go core with SQLite, offline-first, with CLI, Neovim, web and iOS frontends.

2026

VOLUNTEER EXPERIENCE

Lumiere Research Scholar Mentor

SEPT 2025 -

Provided 1-on-1 mentorship to ambitious high school students, guiding them through the creation of independent research papers in mathematics. Supervised the entire research process, from literature review to the authorship of a university-level manuscript, while teaching advanced undergraduate concepts.

UTM Math Club Administrator

Involved in the mathematical community in answering questions, sharing new perspectives, facilitating events, and guiding newer students
(member count: ≈ 3785)

Nov 2020-